

YASEEN KHAN

Aspiring AI / Data Science Researcher | Full-Stack Developer & Applied ML Engineer

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RESEARCH & ACADEMIC PROFILE

University Gold Medalist in Information Technology (CGPA 3.93/4.00, University of Malakand). My academic and applied work centers on machine learning systems for resource-constrained, real-world environments — most recently an association-rule-mining recommendation microservice (DROMS) and a churn-prediction pipeline addressing class imbalance. As lead author, I am currently preparing a research paper on this work for journal submission. I am seeking a research-oriented Master's in Artificial Intelligence / Data Science to formally extend this work, and remain open to research directions across AI, ML, and agentic/automation systems.

RESEARCH INTERESTS

- Open to research across Artificial Intelligence and Machine Learning broadly, with particular interest in: applied ML for recommendation and decision-support systems in low-resource / offline environments; agentic AI, multi-agent orchestration, and intelligent automation; and data robustness and evaluation methodology in applied ML pipelines

EDUCATION

Bachelor of Science in Information Technology 2022 – 2026

University of Malakand, Pakistan

- University Gold Medalist — CGPA 3.93/4.00 (top of cohort)
- Relevant Coursework: Artificial Intelligence, Database Management Systems, Software Engineering, Data Structures & Algorithms, Operations Research, Linear Algebra, Statistics, Digital Image Processing

RESEARCH & PUBLICATIONS

Countering Data Flattening in Seasonal Retail Recommendation *In Preparation — Internal Review Stage*

- Lead author, research paper arising from the DROMS final-year project, addressing recommendation degradation caused by seasonal demand flattening in retail/restaurant transaction data; currently under review by project supervisor and an external professional researcher prior to journal submission.
- Led the study design, and the association-rule-mining and FastAPI ML microservice underlying the analysis, with supervisory guidance from co-author.

ACADEMIC & RESEARCH PROJECTS

DROMS — Digital Restaurant Order Management System *Final Year Project, 2025–2026*

- Designed an on-premise, internet-independent restaurant management ecosystem spanning a web app (Next.js), desktop app (Electron), and mobile app (React Native) — all communicating over a Node.js/Express backend without external internet dependency.
- Implemented Role-Based Access Control (RBAC) across three distinct operational roles — Waiter, Kitchen, and Admin — each with a tailored interface and permission scope.
- Engineered a Python FastAPI ML microservice applying association-rule mining to generate automated combo-deal recommendations from transaction data.
- Findings formed the basis of a research paper currently in preparation (see Publications).

Customer Churn Prediction — End-to-End ML Pipeline *Independent Research Project*

- Built an end-to-end pipeline (Python, Scikit-learn, FastAPI) to predict telecom customer churn, framed around minimizing false negatives given asymmetric business cost of missed at-risk customers.
- Addressed class imbalance and data skew through resampling and F1-optimized logistic regression; achieved >80% predictive accuracy.
- Applied feature-importance analysis to translate model outputs into actionable retention strategy — connecting model performance to decision-making, a core data science skill.

PROFESSIONAL EXPERIENCE

Freelance Full-Stack DeveloperFeb 2024 – Present

Self-employed, remote

- Delivered 10+ commercial web applications for independent clients, including a multi-tenant legal-practice SaaS platform (Lexora) and a scalable LMS (Prosys) supporting bulk multi-course roster operations with sub-50ms cache latency.
- Managed full project lifecycle — requirements gathering, architecture, delivery, and client handover — while mentoring junior developers on the MERN stack.

Digital Marketing & SEO InternFeb 2025 – Aug 2025

Ranking Grow LLC

- Supported SEO optimization and competitive channel analysis workflows, developing applied experience with data-driven digital strategy.

Selected Freelance Projects

- Clinic Appointment Booking System — role-based scheduling platform eliminating booking conflicts (Next.js, TypeScript, Supabase)
- Pharmacy Management System — inventory, sales, and prescription tracking platform for retail pharmacy operations (Next.js, TypeScript, Supabase)
- Local Buy & Sell Marketplace — real-time peer-to-peer inventory and listing platform (Next.js, Supabase)
- Expense Tracker — personal finance dashboard with real-time analytics (Next.js, TypeScript, Supabase)

TECHNICAL SKILLS

- **Programming & ML:** Python, JavaScript/TypeScript, Scikit-learn, FastAPI, association-rule mining, Next.js, React, Node.js
- **AI & Agentic Systems:** LangChain, LangGraph, LangSmith, n8n, FAISS, Chroma DB, Model Context Protocol (MCP)
- **Data & Infrastructure:** PostgreSQL, MySQL, MongoDB, Supabase, Prisma, Git, Vercel
- **Languages:** English (proficiency certified — IELTS/PTE scheduled within 2 months), Urdu, Pashto

HONORS & CERTIFICATIONS

- University Gold Medalist, Information Technology, University of Malakand (2026)
- Advanced Python Programming and Applications — NAVTTC (2026)
- Artificial Intelligence (in progress) — Khyber Pakhtunkhwa IT Board / KPITB (2026)
- Web Development Certification — National Freelancing Training Program, NFTP (2024)
- Graphic Designing and UI/UX — Bano Qabil / Alkhidmat Foundation (2025)

REFERENCES

Dr. Nasir Rashid

Assistant Professor, Department of Computer Science & IT, University of Malakand

Final Year Project Supervisor and co-author | Email: nasir@uom.edu.pk

Dr. Pervez Khan (PhD)

Associate Professor, Department of Computer Science & IT, University of Malakand

Course Instructor — Internet Architecture & Protocols, Web Development, Operating Systems, System and Network Administration | Email: pervezkhan@uom.edu.pk

Dr. Fakhrudin

Assistant Professor, Department of Computer Science & IT, University of Malakand

Course Instructor — Operational Research, Advanced OOP (Java); FYP Part 1 Supervisor | Email: fakhrudin@uom.edu.pk